



UNIVERSITÉ GRENOBLE ALPES / ENSIMAG

Internship Report

Phage-Host Classification Using Conformal Prediction????

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Abstract

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1 Introduction

The rapid rise of antibiotic-resistant bacteria is one of the major challenges that global medicine is facing today. To overcome this issue, researchers are going to alternative solutions, one of which is phage therapy. Phage therapy relies on using the naturally occurring viruses to target and destroy the harmful bacteria.

However, To make this therapy a viable alternative, we first need to understand the basic biology of these viruses. Bacteriophages, or phages are viruses that infect bacterial hosts that and are one of the most abundant entities on the planet. Each phage contains a collection of structural and functional proteins that collectively determine the bacterial strains that the phage can infect - this property is called the host range of the phage.

These phages are highly host-specific: a phage which infects one bacterial host may have very little to no effect on another. This particular property of theirs has naturally generated interest in modelling and applications in the context of..... . Therefore a significant problem in the phage biology is the prediction of the bacterial hosts associated with a given phage. Accurate host-predictions can have applications in

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The host specificity of phages is one of phage therapy's greatest advantage, but at the same time, it also brings a very significant practical challenge. While it allows us to target specific harmful bacteria without destroying the beneficial ones, it also causes problems in finding exactly the appropriate phage for a specific infection. Doing this by only experimental screening can be time-consuming and expensive. This motivates the development of reliable computational models that are capable of predicting these phage-host interactions directly from the protein level information.

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The existing computational tools, however have a very important limitation: they typically only produce point predictions, which are often represented as a scalar confidence score between 0 and 1 with no associated measure of uncertainty. This lack of having an uncertainty quantification is problematic in clinical contexts, where failing to identify a true host could mean missing a life-saving treatment. Consequently, to build trust in automated host prediction, we need a mathematical framework that is capable of quantifying its own predictive uncertainty and provides reliable confidence guarantees.

The primary objective of this report is to address this limitation by developing an uncertainty aware framework for phage-host prediction using **Conformal Prediction**. Conformal prediction provides a distribution-free framework that allows us to transform standard point predictions into prediction sets that are guaranteed to contain the true bacterial host with a user-defined confidence level.

In this project, we build our phage representations by aggregating collections of protein-level functional scores. This process naturally produced high-dimensional, structured score vectors, as different proteins and functions vary in their data availability. These

score vectors also often contain missing or noisy information. To handle these challenges, we explored geometric conformal prediction methods that can construct reliable prediction sets even when dealing with incomplete biological data. As an intermediate step to validate our geometric approach in a controlled environment, we first evaluated the framework on a binary Gram-type classification task before moving on to the full multi-host prediction problem.

2 Background & State of the Art

In this section, we provide the biological, computational and mathematical background and context for the methods developed in this report. We will begin by introducing bacteriophages and the host-prediction problem in Section 2.1, which motivates the need for computational approaches to find a solution. We will then go over the already existing and state-of-the-art computational techniques in Section 2.2, progressing from the classical sequence similarity techniques to the much more efficient and modern protein embeddings techniques. We will also look at the limitations that all the currently existing methods have in common, for example the absence of rigorous distribution-free uncertainty quantification. . In the remaining part of the section, 2.3 and 2.4 we will introduce the mathematical framework of conformal prediction and the score-based aggregation method of in [1], on which we will build our work.

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2.1 Bacteriophages and Host-Prediction Problem

Bacteriophages, commonly referred to as phages, are viruses that specifically infect and replicate within bacteria. They are one of the most abundant biological entities on Earth with an estimated population of 10^{31} in the global biosphere [2], and therefore play a very important role in shaping the microbial communities and evolution. In the recent years the interest in bacteriophages has increased due to the growing threat of antimicrobial resistance and the development of phage therapy as an alternate treatment method to the traditional antibiotics [3].

One of the key characteristics of phages is its host range, i.e. the set of bacterial hosts that the phage is capable of infecting. The host range of a phage is not a fixed property, it is determined by molecular interactions between the phage and the bacterial surface structures during the cycle of infection [2], [4]. Each phage attaches itself to the bacterial host, injects its viral genetic material which are able to replicate inside the host. Therefore, some of the phages only infect a few hosts (have a small host range) as they can only infect a limited number of bacterial strains while there are other which are capable of infecting a broad range of bacterias.

Identifying which phage infects which host, can have several applications both in environmental and clinical studies. In clinical, the host range helps us determine how suitable a phage is for designing the treatment for phage therapy (an alternate method to antibiotics). In environmental studies it is valuable to know the host being infected to understand the ecological interactions between viruses and microbial communities [4] and evolution in the biosphere.

However, host range has also brings its own obstacles. In phage therapy, we need to find precisely the phage that infects the given bacterial pathogen which is the cause of

the infection, and for that we need a fast reliable host prediction method. Traditionally, the method of determining the host range used to depend on laboratory experiments in which candidate phages were tested against a number of bacterial hosts; although this method was able to give accurate reliable results but it was extremely expensive and time consuming. Given the rapid increase in the phage genomes generated by modern technology, the number of possible phage-host interactions has increased astronomically, and this leads us to the need of having a computational prediction method. We need a way which can complement the lab experiment by providing with a shorter list of candidate phages before we move on to experimental validation [5] or even directly infer the host information directly from a given phage.

We can formulate this host prediction problem as follows: let $\mathcal{X} \subseteq \mathbb{R}^d$ be the feature space representing phage representations, and let $\mathcal{H} = \{h_1, \dots, h_N\}$ be the finite label space of possible bacterial hosts.

In a single-host classification setting, we observe random pairs $(X, H) \sim \mathbb{P}_{X,H}$, where $X \in \mathcal{X}$ is a feature vector representing a phage and $H \in \mathcal{H}$ represents the true host. In a general multi-host setting, the target is a non-empty subset of hosts $H^* \subseteq \mathcal{H}$, i.e., $H^* \in \mathcal{P}(\mathcal{H}) \setminus \{\emptyset\}$, where $\mathcal{P}(\mathcal{H}) = 2^{\mathcal{H}}$ denotes the power set of $\mathcal{H}$.

Our primary objective is to construct a set-valued prediction mapping:

$$C : \mathcal{X} \longrightarrow \mathcal{P}(\mathcal{H})$$

which maps an unlabelled test phage $X_{n+1} \in \mathcal{X}$ to a prediction set $C(X_{n+1}) \subseteq \mathcal{H}$ such that it guarantees that the true host is included in the prediction set with a user specified statistical confidence, $1 - \alpha \in (0, 1)$:

$$\mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ denotes the target error probability.

2.2 Current Computational Approaches and Limitations

The problem of predicting which phage infects which host bacteria has been studied computationally for over a decade and the methods have significantly evolved during this time. Prediction methods have progressed from simple rule based alignment to deep representation learning. Here, we will summarize each the primary methodologies of these methods and along with the common limitations they share that motivate us for this work.

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2.2.1 Sequence Similarity and Protein Content Methods

The early host prediction tools rely on sequence alignment between a phage and bacterial genomes, assuming that similarity between the two indicates infection compatibility. Host-Phinder [6] represents each genome g by its normalised k -mer frequency vector $\mathbf{v}(g) \in \mathbb{R}^{4^k}$ and assigns the predicted host through nearest-neighbour search over a reference database $\mathcal{D}_{\text{ref}} = \{(g_i, y_i)\}_{i=1}^N$:

$$\hat{y} = \operatorname{argmax}_{y \in \mathcal{H}} \max_{g_i: y_i=y} \operatorname{sim}(\mathbf{v}(g_{\text{test}}), \mathbf{v}(g_i)).$$

WIsH [7] uses a probabilistic approach, it models each candidate host genome as a variable-order Markov chain with parameters θ_h and then selects the host that maximises the log-likelihood of the new test phage sequence $S = (a_1, \dots, a_L)$:

$$\hat{y} = \operatorname{argmax}_{h \in \mathcal{H}} \ell(S \mid \theta_h), \quad \ell(S \mid \theta_h) = \sum_{i=k+1}^L \log \mathbb{P}(a_i \mid a_{i-k}, \dots, a_{i-1}; \theta_h).$$

The protein-content methods overcome the limitation of whole-genome alignment by decomposing the phage into its constituent proteins. RaFAH [8] clusters phage proteins into M protein clusters via MMseqs2 [9], then it represents each phage by a binary membership vector $\mathbf{x} \in \{0, 1\}^M$, and trains a random forest to map this profile to a distribution over host genera.

$$f_{\text{RF}} : \{0, 1\}^M \longrightarrow \Delta^{|\mathcal{H}|-1}, \quad \text{where } \hat{\mathbf{p}} = f_{\text{RF}}(\mathbf{x}), \quad \sum_{y \in \mathcal{H}} \hat{p}_y = 1$$

This representation is directly relevant to the present work, as our pipeline operates on a related protein-cluster construction, and the data contamination problem discussed in arises from the same source.

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2.2.2 Protein Language Models and Embedding-Based Methods

The current state of the art techniques use protein language models (pLMs), which are neural architectures trained on millions of protein sequences through self-supervised learning [10], [11]. Given a protein sequence $A = (a_1, \dots, a_n) \in \mathcal{A}^n$ over the amino acid alphabet $\mathcal{A}$ of size 20, an encoder Φ_{pLM} produces a residue-level embedding matrix $\mathbf{Z} = \Phi_{\text{pLM}}(A) \in \mathbb{R}^{n \times d}$, it is then mean-pooled to a fixed-length sequence representation:

$$\mathbf{z} = \frac{1}{n} \sum_{i=1}^n \mathbf{Z}_{i,:} \in \mathbb{R}^d.$$

The embedding $\mathbf{z}$ encodes biochemical, structural, and evolutionary information without requiring explicit sequence alignment. Building on ProtT5 [10], tools such as EMPATHI [12] and SUBLYME [13] use these representations for phage protein functional annotation

and metagenomic lysin discovery respectively. These tools are directly involved in this work: the K -dimensional score vector $\mathbf{s} = (s_1(X), \dots, s_K(X))^T \in [0, 1]^K$ that serves as input to our conformal framework is produced by a pipeline built on these annotations, the precise construction of which is described in . For host prediction, protein embeddings have shown substantially improved accuracy over sequence- similarity baselines [14], [15]. So this suggests that the embedding space contains sufficient information for reliable host assignment.

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2.2.3 Shared Limitation: Absence of Uncertainty Quantification

Despite their empirical success of all the methods seen, they all share a fundamental limitation: they produce point predictions (a single host label $\hat{y}$ or a scalar score $\hat{s}(x, y) \in [0, 1]$) without any formal quantification of prediction uncertainty. A score $\hat{s}(x, y) = 0.87$ reflects a relative ranking statistic, not a probability $\mathbb{P}(Y = y \mid X = x)$. And no existing method provides a finite-sample guarantee that the true host is contained in any predicted set with a specified probability.

Existing uncertainty quantification frameworks only offer partial solutions. For example, the Bayesian approaches require strong assumptions on the distributional assumptions that are difficult to justify for biological score data, while post-hoc calibration methods such as Platt scaling provide no finite-sample guarantees on unseen test data [16].

These limitations are consequential in phage therapy, where an incorrectly predicted host can directly harm a patient. This motivates the adoption of conformal prediction [17]. This is a distribution-free framework that wraps any existing scoring model in a statistical construction, and produces prediction sets $C(X) \subseteq \mathcal{H}$ with exact finite-sample coverage guarantees while only requiring exchangeability of calibration and test data.

2.3 Conformal Prediction

Conformal prediction, initially developed by Vovk, Gammerman and Shafer [17] and then later made accessible by Angelopoulos and Bates [16], is a framework for constructing prediction sets that have a coverage guarantee. Unlike the methods seen in 2.2 it can provide us with a set of possible outcomes (not a point prediction), have a mathematical guarantee and yet requires no distributional assumptions on the data, except for the exchangeability condition. Here, we will formalize the framework of conformal prediction and analyze its finite-sample coverage properties.

Definition 2.1 (Exchangeability).

A finite sequence of random variables $Z_1, Z_2, \dots, Z_m$ taking values in a space $\mathcal{Z}$ is exchangeable if, for every permutation σ of the indices $\{1, 2, \dots, m\}$, the joint distribution

of the permuted sequence is identical to that of the original sequence:

$$(Z_1, Z_2, \dots, Z_m) \stackrel{d}{=} (Z_{\sigma(1)}, Z_{\sigma(2)}, \dots, Z_{\sigma(m)})$$

where $\stackrel{d}{=}$ denotes equality in distribution.

Or, equivalently for all measurable sets $A_1, A_2, \dots, A_m \subseteq \mathcal{Z}$ and all permutations $\sigma \in S_m$:

$$\mathbb{P}(Z_1 \in A_1, Z_2 \in A_2, \dots, Z_m \in A_m) = \mathbb{P}(Z_{\sigma(1)} \in A_1, Z_{\sigma(2)} \in A_2, \dots, Z_{\sigma(m)} \in A_m)$$

Independent and identically distributed (i.i.d.) sequences are always exchangeable, but the converse does not always hold. Thus, exchangeability is considered to be a weaker condition than i.i.d., since it requires that the observations have no special ordering. This condition can easily be met by splitting any dataset randomly [1], [17].

2.3.1 Mathematical Setup and Prediction Set

Let $(X, H) \in \mathcal{X} \times \mathcal{H}$ be a random input-label pair, where $\mathcal{X}$ is the feature space that represents our object of interest (in our case it is the phage protein level score vectors) and $\mathcal{H} = \{h_1, \dots, h_N\}$ is the discrete label space (set of candidate hosts).

We assume that we have a calibration dataset, $D_{\text{cal}} = \{(X_1, H_1), \dots, (X_n, H_n)\}$, of n exchangeable data points drawn from an unknown joint distribution, $P_{X,H}$. Then, for a new unlabelled test input, X_{n+1} our aim is to construct a prediction set function $C : \mathcal{X} \rightarrow 2^{\mathcal{H}}$ depending on the data which maps the test input to a subset of candidate labels, i.e., $C(X_{n+1}) \subseteq \mathcal{H}$, such that the true label falls in the set with a defined coverage that is:

$$\mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

where $\alpha \in (0, 1)$ is the significance level. In our experiments we set $\alpha = 0.1$ so that the guarantee is 90%.

Unlike point predictors, $C(X)$ dynamically reflects model uncertainty. It returns a singleton $|C(X)| = 1$ when confident, a multi-label set $|C(X)| \geq 2$ when uncertain between candidates, or an empty set $|C(X)| = 0$ for out-of-distribution inputs. The efficiency of the model can be quantified by the expected set size $\mathbb{E}[|C(X)|]$, an optimal predictor should achieve a coverage $\geq 1 - \alpha$ with the smallest possible set size.

2.3.2 Split conformal prediction

In split conformal prediction, the available data is divided into a training set and a disjoint calibration set, $D_{\text{cal}} = \{(X_i, H_i)\}_{i=1}^n$. The training set is used to fit an underlying scoring model $\hat{f} : \mathcal{X} \times \mathcal{H} \rightarrow \mathbb{R}$. We then define a nonconformity score function $s : \mathcal{X} \times \mathcal{H} \rightarrow \mathbb{R}$ based on $\hat{f}$. This score function quantifies how unusual a pair (x, h) is relative to the

learned patterns, smaller values of $s(x, y)$ indicate stronger conformity between x and label h

Then using the calibration set $\{(X_i, H_i)\}_{i=1}^n$, we evaluate the nonconformity scores:

$$S_i = S(X_i, H_i), \quad \forall i = 1, \dots, n$$

To gurantee that our error rate is less than α we need to compute the conformal calibration quantile, $\hat{q}$, from these calibration scores

Definition 2.2 (Conformal Calibration Quantile).

The conformal calibration quantile q_α is defined as the empirical $\frac{\lceil (n+1)(1-\alpha) \rceil}{n}$ -th quantile of calibration scores $\{S_1, \dots, S_n\}$:

$$q_\alpha := \text{Quantile} \left(\{S_1, \dots, S_n\}; \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right)$$

Equivalently, expressed using the empirical cumulative distribution function:

$$q_\alpha = \inf \left\{ q \in \mathbb{R} : \frac{1}{n} \sum_{i=1}^n \mathbf{1}(S_i \leq q) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right\}$$

Then, for a new input, X_{n+1} the conformal prediction set, $C(X_{n+1})$, is constructed by collecting all the candidate labels in $H \in \mathcal{H}$ whose nonconformity scores do not exceed q_α :

$$C(X_{n+1}) = \left\{ H \in \mathcal{H} \mid s(X_{n+1}, H) \leq q_\alpha \right\} \quad (2.1)$$

The validity of this construction follows from exchangeability. Let $S_{(1)} \leq \dots \leq S_{(n)}$ denote the order statistics of the calibration scores $\{S_1, \dots, S_n\}$, and augment this sequence with $S_{(n+1)} = +\infty$. Then $q_\alpha = S_{(\lceil (n+1)(1-\alpha) \rceil)}$ is the $\lceil (n+1)(1-\alpha) \rceil$ -th order statistic of the augmented set $\{S_1, \dots, S_n, S_{n+1}\}$. Since $(Z_1, \dots, Z_{n+1})$ are exchangeable, the rank of S_{n+1} among the $n+1$ augmented scores is uniformly distributed on $\{1, \dots, n+1\}$. Consequently:

$$\mathbb{P}(S_{n+1} \leq q_\alpha) = \mathbb{P}(\text{rank}(S_{n+1}) \leq \lceil (n+1)(1-\alpha) \rceil) = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1 - \alpha.$$

Theorem 2.3 (Marginal Coverage Guarantee [17], [18]). *If $(X_1, H_1), \dots, (X_{n+1}, H_{n+1})$ are exchangeable, then the prediction set $C(X_{n+1})$ defined in Equation 2.1 satisfies:*

$$\mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

Furthermore, if calibration scores $S_1, \dots, S_n$ are almost surely distinct continuous random

variables, the coverage bound is tight:

$$1 - \alpha \leq \mathbb{P}(H_{n+1} \in C(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1} \quad (2.2)$$

Crucially, theorem 2.3 holds strictly in finite samples without requiring parametric assumptions on $s(\cdot, \cdot)$ or $\mathbb{P}_{X,H}$. While the choice of scoring function s governs efficiency (set size), valid statistical coverage is guaranteed by construction in a single computational pass over D_{cal} [1].

2.3.3 Marginal & Conditional Coverage

Standard conformal prediction satisfies marginal coverage, which means that the $(1 - \alpha)\%$ coverage guarantee holds on average across all possible choices of the calibration set, D_{cal} , and test samples, (X_{n+1}, H_{n+1}) :

$$\mathbb{E}_{D_{\text{cal}}, (X_{n+1}, H_{n+1})} [\mathbb{I}(H_{n+1} \in C(X_{n+1}))] = \mathbb{P}(H_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$$

However, the marginal coverage does not ensure the conditional coverage, given specific features ($X = x$), or specific classes ($H = h$). So, for a dataset that has class imbalance, which is very common in phage-host interaction, a marginal model can satisfy the coverage guarantee by potentially over-covering a majority class (100%) and under-covering a small class (0%) [1], [19]. For instance, if $\mathcal{H} = \{-1, 1\}$ with class proportion $\mathbb{P}(H = -1) = 0.9$, a predictor achieving 100% coverage on $H = -1$ and 0% coverage on $H = 1$ still attains an overall 90%. In phage-host classification, such conditional under-coverage on minority host may be a problem.

While feature-conditional coverage ($\mathbb{P}(H \in C(X) \mid X = x) \geq 1 - \alpha$) is mathematically impossible to achieve in finite samples without strong parametric assumptions [1], [17], class-conditional coverage (Mondrian conformal prediction) is exact and valid in finite samples. We partition D_{cal} into N disjoint subsets $D_{\text{cal}}^{(h)} = \{(X_i, H_i) \in D_{\text{cal}} : H_i = h\}$ and evaluate separate class-specific quantiles $q_{\alpha}^{(h)}$, enforcing exact coverage per class:

$$\mathbb{P}\left(H_{n+1} \in C(X_{n+1}) \mid H_{n+1} = h\right) \geq 1 - \alpha, \quad \forall h \in \mathcal{H} \quad (2.3)$$

Because exchangeability holds within each class-conditional partition $D_{\text{cal}}^{(h)}$, evaluating per-class quantiles preserves exact finite-sample validity across every candidate host, as formalised in Section

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2.4 Conformal Set Aggregation (CSA)

The standard split conformal framework in 2.3.2 takes a scalar non-conformity score $S(x, h) \in \mathbb{R}$. In this report, the prediction pipeline produces a K -dimensional score vector of scores $s \in [0, 1]^K$ for each phage, where K is the number of distinct functional categories. We may lose the geometric structure of the score distribution if we aggregate this vector to a scalar value before applying conformal prediction. For such settings, Ochoa Rivera, Patel and Tewari [1] have proposed a score-based aggregation (CSA) method for conformal prediction.

2.4.1 Ensemble Score Vectors

Let $\mathcal{S} \subseteq [0, 1]^K$ be the score space and let $s = (s_1, \dots, s_K) \in \mathcal{S}$ be the vector of scores produced by the K models, for each phage. Now, the CSA method treats the entire vector as the object of inference rather than reducing it into a single value. The main idea is to define a region, called an envelope, $E \subseteq [0, 1]^K$. This envelope consists of the region where the score vectors of the true hosts for that particular phage are concentrated.

Mathematically, let $\{\phi_h\}_{h \in \mathcal{H}}$ be a family of label-dependent maps $\phi_h : [0, 1]^K \rightarrow [0, 1]^K$ such that, when h is the true label, the transformed score vector $\phi_h(\mathbf{s})$ concentrates near the origin in $\mathbb{R}_+^K$. The CSA method constructs an envelope $E \subseteq \mathbb{R}_+^K$ as the intersection of M half-spaces induced by unit directions $\{\mathbf{u}_m\}_{m=1}^M$ and quantile thresholds $\{\tilde{q}_m\}_{m=1}^M$:

$$E = \bigcap_{m=1}^M \{\mathbf{s} \in \mathbb{R}_+^K : \langle \mathbf{s}, \mathbf{u}_m \rangle \leq \tilde{q}_m\}. \quad (2.4)$$

By construction, E is convex. Membership of a transformed score in the scaled envelope $t \cdot E$ is tested coordinate-wise along each direction. The exact transformation ϕ_h used in this work is formalized in Section .

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2.4.2 The CSA Framework

The core innovation of the CSA method is in the calibration process. It splits the calibration data into two subsets which have different roles. This is necessary for the exchangeability argument that provides us with the coverage guarantee. If we used the same calibration points to construct both, we would get overfitting which would introduce bias which would invalidate the distribution free coverage guarantee [1].

- (i) **Shape Discovery Set (S_1):** Used to model the spatial distribution of scores and learn the geometric shape, orientation, and boundaries of the envelope $E \subset \mathbb{R}^K$.
- (ii) **Size Scaling Set (S_2):** Used to determine the scalar scaling parameter $\hat{t} \in \mathbb{R}^+$. $\hat{t}$ inflates or deflates E until exact empirical coverage $1 - \alpha$ is attained over S_2 :

$$\hat{t} = \inf \left\{ t > 0 \mid \frac{1}{|S_2|} \sum_{\mathbf{s} \in S_2} \mathbf{1}(\phi_H(\mathbf{s}) \in t \cdot E) \geq \frac{\lceil (|S_2| + 1)(1 - \alpha) \rceil}{|S_2|} \right\}$$

This $\hat{t}$ is the smallest scaling factor such that the scaled envelope $\hat{t} \cdot E$ achieves the required $1 - \alpha$ coverage. The scaled envelope is defined linearly as:

$$\hat{t} \cdot E = \{t \cdot x : x \in E\}, \quad \hat{t} > 0$$

In the original CSA paper [1], finding $\hat{t}$ is presented as a root-finding problem solved through iterative numerical optimization or binary search over candidate scaling values t . At test time, a candidate label h is included in the prediction set $C(\mathbf{s}_{n+1})$ if and only if its transformed score vector falls inside the scaled envelope:

$$C(\mathbf{s}_{n+1}) = \{h \in \mathcal{H} : \phi_h(\mathbf{s}_{n+1}) \in \hat{t} \cdot E\}.$$

2.4.3 Limitations of CSA in Classification

This CSA framework was originally developed and analyzed in a regression setting [1], and has some major limitations when it is applied to classification problems:

- (i) **Convexity of Envelope:** In [1], the conformal envelope E is constructed as an intersection of half-spaces and is therefore convex. But in classification, the score vectors for different classes concentrate in separate regions in the $[0, 1]^K$ score space. Therefore, if we force a convex envelope across the regions, we will create a region that covers a large area of empty score space in between, which would inflate the volume of the envelope. This would lead to an over conservative and uninformative prediction sets, $|C(\mathbf{s})|$, as they would be really large.
- (ii) **Calibration Inefficiency:** Originally the method uses an iterative numerical optimization or a binary search to find the optimal scaling factor, $\hat{t}$ over S_2 . In cases when the score space is high dimensional, the iterative approach can be really expensive computationally.
- (iii) **Only Marginal Guarantee:** The scaling calculations in the original method only provide us with a marginal guarantee. This can lead to poor performance on the biological datasets, which are imbalances

These limitations directly motivate our methodological contributions in 3. This includes designing non-convex envelopes, deriving closed-form analytical solutions for per-point τ -values to enable one-pass calibration 3, and integrating class-conditional thresholds to guarantee reliable host-range predictions.

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3 Methodology

In this section, we will develop the mathematical framework used to construct prediction sets with guaranteed finite-sample phage-host classification. Section 3.1 formalises the prediction task and addresses missing data in our score vector. Then, building on the limitations of the CSA method identified in Section 2.4.3, Section 3.2 explains why non-convex envelopes are better suited for classification, and Section 3.3 defines the label-dependent nonconformity transformation used throughout this report to center valid score distributions near the origin. Section 3.4 then presents the three envelope geometries evaluated in this work, Radial, Strip, and Collapsed 1D, and derives the calibration scaling factor $\hat{t}$.

3.1 Problem Formalization

We will now apply the general conformal prediction setup from Section 2.3 to our host prediction task. Recall from Section 2.1 that $\mathcal{H} = \{h_1, \dots, h_N\}$ represents the discrete set of N candidate bacterial hosts. Our goal is to build a function $C : \mathcal{X} \rightarrow \mathcal{P}(\mathcal{H})$ that outputs a set of hosts while maintaining the coverage guarantee in Equation (2.1)

In our work, the feature space, $\mathcal{X}$, does not use the raw phage sequences directly. Instead it consists of fixed-length score vectors (K dimensional) generated by the protein annotation process introduced in Section 2.2.2. So each score s_k corresponds to a specific functional protein category (e.g., capsid, tail, lysin), where K is the total number of categories. A specific phage genome may lack proteins in different functional categories, therefore the score vector may have structural missingness:

$$\mathbf{s} = (s_1, \dots, s_K)^T \in ([0, 1] \cup \{\text{NA}\})^K.$$

Here, $s_k = \text{NA}$ is the structural absence, which means that for category k no annotated protein was identified in the phage sequence, rather than a zero-evidence score or missingness caused by random measurement noise. This value is missing by design, not due to noise, and hence we need to handle it explicitly rather than simply imputing it.

To retain the mathematical guarantee without imputations, we attach a missingness mask, $\mathbf{m}$, to every score vector. We represent each score observation as a paired vector $(\mathbf{s}, \mathbf{m}) \in ([0, 1] \cup \{\text{NA}\})^K \times \{0, 1\}^K$:

$$\mathbf{m} = (m_1, \dots, m_K)^T \in \{0, 1\}^K, \quad m_k = \mathbf{1}(s_k \neq \text{NA}).$$

The total number of observed functional dimensions for a given phage is given by the L_1 norm of the mask, $\|\mathbf{m}\|_1 = \sum_{k=1}^K m_k$. Then we have $\mathbf{s}_{\mathbf{m}} \in [0, 1]^{\|\mathbf{m}\|_1}$, the sub-vector restricted strictly to the observed dimensions where $m_k = 1$. All prediction functions and

nonconformity scoring operators developed in this report are defined over the product space of pairs $(\mathbf{s}, \mathbf{m})$, rather than $\mathbf{s}$ alone. This ensures our nonconformity scores and envelope tests remain mathematically valid for any missingness pattern, not just fully observed data, $\mathbf{m} = \mathbf{1}$.

We apply this framework to two host prediction tasks:

- (i) **Gram-type classification** ($N = 2$): $\mathcal{H} = \{\text{gram-neg}, \text{gram-pos}\}$, evaluated over K functional category dimensions.
- (ii) **Host-level classification** ($N > 2$): $\mathcal{H}$ represents the full set of bacterial hosts in our dataset. For each host $h \in \mathcal{H}$, the score space is decomposed into host-specific sub-vectors of dimension K_h , and an individual envelope E_h is fitted independently on training phages with true host h .

In practice, host-level classification is not evaluated over the full host space $\mathcal{H}$ at test time. We first obtain a gram-type prediction set $C_{\text{gram}}(\mathbf{s}) \subseteq \{\text{gram-neg}, \text{gram-pos}\}$ from the procedure of Section , and use it to restrict the candidate host set to

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$$\hat{\mathcal{H}}(\mathbf{s}) = \bigcup_{g \in C_{\text{gram}}(\mathbf{s})} \mathcal{H}_g \subseteq \mathcal{H},$$

where $\mathcal{H}_g \subset \mathcal{H}$ denotes the set of hosts of gram-type g . Let $s_h \in ([0, 1] \cup \{NA\})^{K_h}$ denotes the restriction of s to the K_h score dimensions associated with host h . The host-level prediction map of this report is therefore the composition

$$C_{\text{host}}(\mathbf{s}, \mathbf{m}) = \{h \in \hat{\mathcal{H}}(\mathbf{s}) \mid \phi_h(\mathbf{s}_h) \in \hat{t}_h \cdot E_h\},$$

rather than a single-stage test over all of $\mathcal{H}$. This cascade reduces the number of per-host envelope evaluations required at test time, at the cost of making C_{host} dependent on the upstream gram-type prediction; we will see the implications of this dependency in Section .

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Given a calibration set $D_{\text{cal}} = \{(\mathbf{s}_i, \mathbf{m}_i, H_i)\}_{i=1}^n$ of exchangeable samples, and an unlabelled test sample $(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})$ drawn exchangeably from an unknown joint probability distribution. Our goal is to construct a set-valued prediction mapping $C : ([0, 1] \cup \{NA\})^K \times \{0, 1\}^K \rightarrow \mathcal{P}(\mathcal{H})$ at significance level $\alpha \in (0, 1)$ such that:

$$\mathbb{P}(H_{n+1} \in C(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})) \geq 1 - \alpha.$$

At the same time, to maximize the predictive efficiency, we want to minimize the expected set size $\mathbb{E}[|C(\mathbf{s}, \mathbf{m})|]$. The sets $C(\mathbf{s}_{n+1}, \mathbf{m}_{n+1})$ as defined above may be empty when no candidate label's transformed score falls within the calibrated envelope. The two prediction tasks in this report resolve this case differently 3.4.

3.2 Non-Convex Envelopes in Classification

Section 2.4.3 identified envelope convexity as one of the main limitations of standard CSA method [1] when applied to classification. While convex envelopes naturally model continuous target spaces in regression, classification score vectors concentrate in disjoint, class-specific clusters across the score space. Therefore, it leads to the envelope being stretched over regions of empty spaces, making the prediction sets unnecessarily large. Here, we will describe the alternate method that we have used in this report.

Formally, let $\mathcal{C}_h \subseteq \mathbb{R}_+^K$ denote the support of the transformed NC vector $\phi_h(\mathbf{s})$ for true label $H = h \in \mathcal{H}$. In a multi-class classification, these supports are strictly pairwise separated:

$$\mathcal{C}_h \cap \mathcal{C}_{h'} = \emptyset, \quad \text{dist}(\mathcal{C}_h, \mathcal{C}_{h'}) > 0, \quad \forall h \neq h' \in \mathcal{H}. \quad (3.1)$$

A single convex envelope $E_{\text{conv}} = \text{conv}(\bigcup_{h \in \mathcal{H}} \mathcal{C}_h)$ must contain every line segment connecting points in $\mathcal{C}_h$ and $\mathcal{C}_{h'}$. These segments inevitably traverse unpopulated space. Letting $\text{gap} := E_{\text{conv}} \setminus \bigcup_{h \in \mathcal{H}} \mathcal{C}_h$, the Lebesgue volume decomposes as:

$$\text{Vol}(E_{\text{conv}}) = \sum_{h \in \mathcal{H}} \text{Vol}(\mathcal{C}_h) + \text{Vol}(\text{gap}), \quad \text{where } \text{Vol}(\text{gap}) > 0. \quad (3.2)$$

Test points landing are obviously included in the prediction set, $|C(\mathbf{s})|$, and they make them larger without improving coverage. This inefficiency scales with the number of classes $N = |\mathcal{H}|$, making shared convex hulls particularly sub-optimal for host-level classification with multiple classes ($N > 2$).

Also very importantly, dropping convexity does not violate the statistical validity: the marginal coverage guarantee holds for any measurable envelope $E \subseteq \mathbb{R}_+^K$ by exchangeability [1]. To ensure the scaling function $\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\}$ yields a single, well-defined scalar threshold, E needs only to be star-shaped (Definition 3.1) with respect to the origin:

$$\mathbf{x} \in E \implies \lambda \mathbf{x} \in E, \quad \forall \lambda \in [0, 1].$$

Definition 3.1 (Star-Shaped Set).

A set $E \subseteq \mathbb{R}^K$ is said to be *star-shaped with respect to the origin* $0 \in \mathbb{R}^K$ if for every point $x \in E$ and every scalar $\lambda \in [0, 1]$, the line segment connecting the origin to x lies entirely within E :

$$\lambda x \in E \quad \forall x \in E, \forall \lambda \in [0, 1].$$

Star-shapedness guarantees that $\{t > 0 : \mathbf{s} \in t \cdot E\}$ forms a continuous interval, preserving simple single-pass calibration while allowing the envelope to tightly hug non-convex cluster geometries.

3.3 Nonconformity Score Transformation

The raw score vectors are not directly comparable across candidate labels and cannot be used directly as nonconformity measures. Recall from Section 2.4.1 that the CSA framework requires a label-dependent transformation $\phi_h : ([0, 1] \cup \{\text{NA}\})^K \times \{0, 1\}^K \rightarrow (\mathbb{R}_+ \cup \{\text{NA}\})^K$ that maps candidate hypotheses into a nonconformity score space centered near the origin.

The raw functional category scores $s_k \in [0, 1]$ quantify evidence in favor of functional compatibility, therefore a raw score close to 1 reflects strong infection compatibility (low nonconformity), whereas a score near 0 reflects high nonconformity. Thus, the mapping is defined such that when a candidate hypothesis $h \in \mathcal{H}$ corresponds to the true host label, the resulting vector $\mathbf{v} = \phi_h(\mathbf{s}, \mathbf{m})$ concentrates near $\mathbf{0} \in \mathbb{R}_+^K$, whereas incorrect hypotheses yield vectors situated significantly further from the origin. We now formalize ϕ_h for both classification settings while explicitly preserving structural missingness via the mask vector $\mathbf{m}$.

To map all the score vectors into one unified nonconformity space near the origin, we define label-dependent transformations ϕ_h . Formally, for a single score dimension $s_k \in [0, 1]$ with the binary mask $m_k \in \{0, 1\}$, we define the general coordinate wise mapping:

$$\phi_h(s_k, m_k) = \begin{cases} |s_k - c_h| & \text{if } m_k = 1, \\ \text{NA} & \text{if } m_k = 0, \end{cases} \quad c_h \in \{0, 1\}, \quad (3.3)$$

where the c_h tells us whether a low ($c_h = 0$) or high ($c_h = 1$) raw score provides the evidence for hypothesis h . $s_k \in [0, 1]$, (3.3) can be simplified to $\phi_h(s_k) = s_k$ when $c_h = 0$, and $\phi_h(s_k) = 1 - s_k$ when $c_h = 1$, this ensures that $\phi_h(s_k) \rightarrow 0$ whenever $s_k \rightarrow c_h$.

3.3.1 Gram Type Classification Nonconformity Mapping

In the binary Gram-type classification, $\mathcal{H} = \{\text{gram-neg}, \text{gram-pos}\}$, all the K functional score models measure a single Gram-positive evidence scale. Empirically, true Gram-negative phages have low raw scores (mean ≈ 0.06 – 0.08), whereas true Gram-positive phages exhibit high raw scores (mean ≈ 0.63 – 0.81).

Since the scale is shared, the c_h must differ by hypothesis: $c_{\text{gramneg}} = 0$ and $c_{\text{grampos}} = 1$. This gives us the elementwise transformation for dimension $k \in \{1, \dots, K\}$:

$$v_k = (\phi_h(\mathbf{s}, \mathbf{m}))_k = \begin{cases} s_k & \text{if } h = \text{gram-neg and } m_k = 1, \\ 1 - s_k & \text{if } h = \text{gram-pos and } m_k = 1, \\ \text{NA} & \text{if } m_k = 0. \end{cases} \quad (3.4)$$

Using (3.4), we get a high nonconformity score if we test an incorrect hypothesis, and

the scores for the true hypothesis are mapped near the origin $\mathbf{0} \in \mathbb{R}_+^K$. This allows both classes to be calibrated against a shared reference envelope $E \subset \mathbb{R}_+^K$.

3.3.2 Host Level Classification Nonconformity Mapping

In the host level setting ($N > 2$), the candidate labels do not share the scores. Instead, each candidate host $h \in \mathcal{H}$ is evaluated on a host-specific sub-vector $\mathbf{s}_h \in ([0, 1] \cup \{\text{NA}\})^{K_h}$. This measures the direct functional compatibility with host h .

A high raw score $s_{h,k}$ indicates high likelihood of infection, $c_h = 1$ for all hosts, using this we can get the following host-specific mapping across $k \in \{1, \dots, K_h\}$:

$$v_{h,k} = (\phi_h(\mathbf{s}_h, \mathbf{m}_h))_k = \begin{cases} 1 - s_{h,k} & \text{if } m_{h,k} = 1, \\ \text{NA} & \text{if } m_{h,k} = 0. \end{cases} \quad (3.5)$$

Unlike the Gram classification task, in host level prediction we construct N separate host envelopes $E_h \subset \mathbb{R}_+^{K_h}$ which are fitted independently on calibration phages with true label $H = h$. At test time, $\mathbf{s}_h$ is transformed using (3.5) and tested against E_h , this avoids forcing unrelated host signatures into a single shared coordinate system.

Both transforms are strictly monotone fixed maps that require no parameter estimation from data. Therefore, they do not have sample dependencies that could break the exchangeability argument (which is necessary for the coverage guarantee, Theorem 2.3). The missingness masks $\mathbf{m}$ propagate unchanged as NA values, and the envelope constructions of Section 3.4 operate directly on this representation.

3.4 Envelope Construction

Here we will present the three envelope geometries that we have evaluated in this report. Each method follows the two-stage calibration procedure: the shape-discovery set S_1 determines the geometric form of the envelope, and the size scaling set S_2 determines the scaling factor $\hat{t}$. All three envelopes are star-shaped with respect to the origin, so the per-point scaling function

$$\tau(\mathbf{s}) = \inf\{t > 0 : \mathbf{s} \in t \cdot E\} \quad (3.6)$$

is well-defined as a single scalar for every $\mathbf{s} \in \mathbb{R}_+^K$. For each envelope type, we derive a closed form expression for $\tau(\mathbf{s})$ that eliminates the need for iterative search. The scaling factor $\hat{t}$ is then computed using class conditional quantile aggregation over S_2 , as detailed in Section 3.5.

3.4.1 Collapsed 1D Envelope Method

The simplest envelope construction method just collapses the K -dimensional (or K_h -dimensional) NC vector to a scalar value by taking the mean over the over observed dimensions,

$$\bar{s} = \frac{1}{|\mathbf{m}|_1} \sum_{k:m_k=1} v_k,$$

and calibrates a single quantile on S_1 :

$$\tilde{q} = \text{Quantile}_{1-\alpha}(\{\bar{s}_i\}_{i \in S_1}), \quad \tau(\mathbf{s}) = \frac{\bar{s}}{\tilde{q}}, \quad E = \{\mathbf{s} : \bar{s} \leq \tilde{q}\}.$$

This envelope is star-shaped (Section 3.2), it is unaffected by the dimensionality K , but at the same time is also discards all the geometric information that the K dimensions provide us with. Any two vectors with the same mean are treated identically regardless of the individual coordinates.

3.4.2 Radial Envelope Method

The Radial envelope method retains the directional structure of the scores by calibrating a boundary radius as a function of direction on the unit sphere which is restricted only to the positive orthant (since all NC values are non-negative). In this method, we draw M directional vectors $\{\mathbf{u}_m\}_{m=1}^M$ uniformly over the orthant by sampling $\mathbf{v}_m \sim |\mathcal{N}(0, I_K)|$ coordinate-wise and normalizing: $\mathbf{u}_m = \mathbf{v}_m / \|\mathbf{v}_m\|$.

For each S_1 point $\mathbf{s}_i$, over the observed dimensions only, we compute its magnitude and unit direction, $r_i = \|\mathbf{s}_{i,\mathbf{m}_i}\|$, $\mathbf{d}_i = \mathbf{s}_{i,\mathbf{m}_i}/r_i$. For each sample direction $\mathbf{u}_m$, we collect n_m calibration points within an angular sector of half-width $\delta = 30^\circ$ (where $\mathbf{d}_i \cdot \mathbf{u}_m \geq \cos \delta$). The directional radius $\tilde{q}_m$ is then calibrated as:

$$\tilde{q}_m = \begin{cases} \text{Quantile}_{1-\alpha}(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos \delta\}) & \text{if } n_m \geq n_{\min}, \\ w_m \cdot \text{Quantile}_{1-\alpha}(\{r_i : \mathbf{d}_i \cdot \mathbf{u}_m \geq \cos \delta\}) + (1 - w_m) \cdot \tilde{q}_{\text{marg}} & \text{if } 0 < n_m < n_{\min}, \\ \tilde{q}_{\text{marg}} & \text{if } n_m = 0, \end{cases}$$

where n_m is the number of S_1 points within the band of direction m , $n_{\min}$ a minimum-support threshold, $w_m = n_m/n_{\min}$, and $\tilde{q}_{\text{marg}} = \text{Quantile}_{1-\alpha}(\{r_i\}_{i \in S_1})$ the marginal fall-back radius. This blend enables us to avoid a hard cutoff between sufficient and insufficient local support (number of points in that region).

To ensure that we get a smooth boundary, we compute the radius at an arbitrary direction $\mathbf{d}$ as a directional weighted average of all M calibrated radii:

$$r(\mathbf{d}) = \sum_{m=1}^M w_m(\mathbf{d}) \tilde{q}_m, \quad w_m(\mathbf{d}) = \frac{\exp(\kappa \mathbf{d} \cdot \mathbf{u}_m)}{\sum_{m'=1}^M \exp(\kappa \mathbf{d} \cdot \mathbf{u}_{m'})}, \quad (3.7)$$

where $\kappa > 0$ controls the angular smoothness of the interpolation across directions (larger κ approaches nearest-direction assignment). This makes the envelope boundary a continuous function of direction rather than a piecewise constant one. The scaling function follows directly from this:

$$\tau(\mathbf{s}) = \frac{\|\mathbf{s}_{\mathbf{m}}\|}{r(\mathbf{d})}, \quad \mathbf{d} = \frac{\mathbf{s}_{\mathbf{m}}}{\|\mathbf{s}_{\mathbf{m}}\|}, \quad E = \{\mathbf{s} : \|\mathbf{s}_{\mathbf{m}}\| \leq r(\mathbf{d})\}.$$

As established in Section 3.2, this construction is star-shaped by design. This is because r depends only on direction and not on magnitude, $\mathbf{s} \in t \cdot E$ for some t implies $\lambda \mathbf{s} \in t \cdot E$ for all $\lambda \in [0, 1]$.

3.4.3 Strip Envelope Method

The Strip envelope defines conditional bounds for each target dimension i based on the bin occupied by another dimension j . We partition the range of dimension j into NB equal-width bins. For each bin n containing $n_{j,n}$ calibration points, the upper limit along dimension $i \neq j$ is calibrated as:

$$\text{limits}[j, i, n] = \begin{cases} \text{Quantile}_{1-\alpha}(\{s_{:,i} : s_{:,j} \in \text{bin } n\}) & \text{if } n_{j,n} \geq n_{\min}, \\ w_{j,n} \cdot \text{Quantile}_{1-\alpha}(\{s_{:,i} : s_{:,j} \in \text{bin } n\}) + (1 - w_{j,n}) \cdot \tilde{q}_i & \text{if } 0 < n_{j,n} < n_{\min}, \\ \tilde{q}_i & \text{if } n_{j,n} = 0, \end{cases}$$

where $\tilde{q}_i$ is the marginal $(1 - \alpha)$ quantile for dimension i over S_1 , and $w_{j,n} = n_{j,n}/n_{\min}$ shrinks underpopulated bins toward it (it mirrors the blending rule of the radial envelope).

A windowed monotonicity pass then enforces $\text{limits}[j, i, n] \geq \text{limits}[j, i, n']$ for $n' \in (n, n + w]$ within a fixed window w . This is done to ensure that a bin's limit cannot be undercut by a bin immediately ahead of it in the same conditioning dimension. The monotonicity is necessary to ensure that a phage with a smaller value (more confident) in the conditioning dimension j , does not receive a tighter bound in dimension i than a phage with a larger value in j , which makes sure that the conditional envelope is logically correct.

$$\text{limits}[j, i, n] \leftarrow \max_{n' \in [n, \min(n+w, NB)]} \text{limits}[j, i, n']$$

The scaling function $\tau(\mathbf{s})$ aggregates the worst-case ratio over all conditioning pairs (j, i) :

$$\tau(\mathbf{s}) = \max_{j \neq i} \frac{s_i}{\text{limits}[j, i, \text{bin}_j(\mathbf{s})]}, \quad E = \{\mathbf{s} : \forall j \neq i, s_i \leq \text{limits}[j, i, \text{bin}_j(\mathbf{s})]\}.$$

Missing dimensions ($m_k = 0$) are handled pairwise:

- **Conditioning dimension j missing** ($m_j = 0$): The bin index along j is undefined, so the evaluation falls back to the normalized marginal ratio $s_i/\tilde{q}_i$.

- **Target dimension i missing:** Its value s_i is replaced by its marginal quantile $\tilde{q}_i$ before computing the ratio.
- **Both missing:** The pair contributes no constraint (ratio = 0), since absence of evidence should not lead to a false rejection.

3.5 $\hat{t}$ Calculation

3.5.1 τ Computation

For any test point $\mathbf{s} \in \mathbb{R}^K$, we define the inclusion threshold function $\tau(\mathbf{s})$ (defined in 3.6) as the minimum scaling factor required for the scaled envelope $t \cdot E$ to contain $\mathbf{s}$:

Given the computed τ values over the scaling set S_2 , the calibrated scaling factor $\hat{t}$ is selected as the $(1 - \alpha)$ empirical quantile:

$$\hat{t} = \tau_{(k)} \quad \text{where} \quad k = \lceil (|S_2| + 1)(1 - \alpha) \rceil$$

Here, $\tau_{(k)}$ denotes the k -th order statistic (the k -th smallest value) of $\{\tau(\mathbf{s}_i)\}_{\mathbf{s}_i \in S_2}$.

Unlike the original CSA method, which relies on a computationally expensive iterative binary search over candidate t values, our closed form expressions allow $\tau(\mathbf{s})$ to be evaluated directly for each point in a single pass over S_2 . This eliminates the iterative search entirely, which reduces the calibration computational complexity from $O(|S_2| \cdot N_{\text{iter}})$ to $O(|S_2| \log |S_2|)$.

3.5.2 Class Specific $\hat{t}$ Calculation

When the dataset has class imbalance (like our dataset), a single global scaling factor $\hat{t}$ can cause under coverage for complex classes and over coverage for simpler ones. To enforce exact class conditional coverage $1 - \alpha$, we partition S_2 into class-specific calibration sets $S_2^{(c)} = \{\mathbf{s} \in S_2 : h = c\}$ for each label $c \in \mathcal{H}$.

For each class c , the class-specific scaling threshold $\hat{t}_c$ is calculated independently on $S_2^{(c)}$:

$$\hat{t}_c = \tau_{(k_c)}^{(c)}, \quad k_c = \lceil (|S_2^{(c)}| + 1)(1 - \alpha) \rceil. \quad (3.8)$$

The two prediction tasks in this report differ in how envelopes are constructed, which determines how these per class thresholds $\hat{t}_c$ are applied:

Gram-Type Classification: We only fit a single envelope, E on the shared shape discovery set S_1 containing phages of both classes. If we evaluate a single pooled threshold over S_2 , it would only yield only marginal coverage. So, we compute the class specific quantiles $\hat{t}_{\text{gram-neg}}$ and $\hat{t}_{\text{gram-pos}}$ using (3.8) and scale the shared envelope by their maxi-

mum:

$$\hat{t} = \max(\hat{t}_{\text{gram-neg}}, \hat{t}_{\text{gram-pos}}). \quad (3.9)$$

Since $\hat{t} \geq \hat{t}_c$ for both classes, scaling E by $\hat{t}$ satisfies both coverage requirements simultaneously, guaranteeing class-conditional validity $\mathbb{P}(\mathbf{s} \in \hat{t} \cdot E \mid H = c) \geq 1 - \alpha$ for $c \in \{\text{gram-neg}, \text{gram-pos}\}$.

Host-Level Classification: In host-level classification, each candidate host $h \in \mathcal{H}$ has a dedicated envelope E_h fitted exclusively on phages with true host h (Section 3.1). Because its calibration set $S_2^{(h)}$ is already single-class, $\hat{t}_h$ is computed directly via 3.8 without taking a maximum across hosts. Each scaled envelope $\hat{t}_h \cdot E_h$ is validated independently, admitting a candidate host h if $\phi_h(\mathbf{s}_h) \in \hat{t}_h \cdot E_h$.

3.5.3 Resolving the Empty Prediction Set

In section 3.1 we noted that $C(\mathbf{s}, \mathbf{m})$ may be empty when no candidate label’s transformed score falls inside $\hat{t} \cdot E$. In both our tasks we resolve this issue differently.

For gram-type classification, we resolve an empty set conservatively by returning both labels, $C(\mathbf{s}) = \mathcal{H}$, rather than an arbitrary tie-break — since $N = 2$, this is the uninformative but always-valid choice. On the other hand, for host-level classification, where N can be large and returning all of $\hat{\mathcal{H}}(\mathbf{s})$ would be uninformative, an empty set is instead resolved by forcing a singleton at the host with the closest τ value:

$$C(\mathbf{s}) = \left\{ \underset{h \in \hat{\mathcal{H}}(\mathbf{s})}{\operatorname{argmin}} \tau_h(\mathbf{s}_h) \right\}.$$

Both are applied only when the primary rule of Equation (2.1) yields $\emptyset$. The coverage guarantee of Theorem 2.3 remains unaffected as it is established on the primary construction over $\mathcal{H}$ and empty outcomes are, by definition, cases where that guarantee already requires some label to be returned to satisfy the $1 - \alpha$ bound in expectation. Because set expansion and nonempty fallback options strictly enlarge $C(\mathbf{s})$, the lower bound $\mathbb{P}(H \in C(\mathbf{s})) \geq 1 - \alpha$ trivially holds. Consequently, the reported `empty_rate` for both tasks is strictly zero by rule rather than an absence of out of envelope points. To preserve visibility of these points, Section 5 reports the pre resolution `forced_rate` separately.

4 Data and Preprocessing Pipeline

In this section, we describe the dataset architecture 4.1, the mathematical aggregation operators used to combine protein-level functional annotations into phage-level score vectors 4.5, and the masking strategy developed to eliminate bias caused by cross-fold protein cluster contamination 4.7.

4.1 Data Description and Structural Missingness

Our dataset comprises of $N_{\text{prot}} = 1\,853\,074$ protein sequences extracted from 18 474 unique bacteriophages. The base models use the EMPATHI pipeline (Section 2.2.2) to predict functional annotations across categories such as lysin, tail sheath, portal protein, and capsid [5]. An entry $s_{p,m} \in [0, 1]$ is the model’s predicted probability that protein p carries function m . In this work we treat these scores as fixed features in our conformal framework without retraining the underlying classifiers.

Our pipeline operates on four primary data structures:

- (i) **Protein Score Matrix:** Matrix of raw model predictions $\mathbf{S} \in ([0, 1] \cup \{\text{NA}\})^{N_{\text{prot}} \times M}$, where entry $s_{p,m}$ denotes the predicted probability that protein p carries function m . For the Gram-type task, we evaluate the M functional categories aggregated across all hosts; for the host-level task, $M = 2\,160$ host-function pairs.
- (ii) **Phage Metadata:** Mappings linking each protein ID to its accession identifier, host genus label $H \in \mathcal{H}$ ($N = 54$ genera), host Gram-type $G \in \{\text{gram-neg}, \text{gram-pos}\}$, and protein cluster identifier (**pc**).
- (iii) **Cross-Validation Partition:** Integer fold assignments $f \in \{0, 1, 2, 3, 4\}$ assigned at the genome level, they ensure that all proteins from a given phage share the same fold.
- (iv) **Protein Split Mask Matrix:** A binary assignment matrix mapping each protein to its evaluation fold across all host-function models.

The dataset contains some structural missingness: approximately 30.6% to 36.5% of protein category scores are missing (NA) because phages possess only a subset of functional categories. An entry $s_{p,m} = \text{NA}$ indicates that protein p was not annotated with function m . This is deterministic rather than stochastic: a capsid protein cannot carry a tail-fiber score, and an unannotated cluster does not provide any predictions.

Formally, we can say that there exists a deterministic missingness mask $\mathbf{M} \in \{0, 1\}^{N_{\text{prot}} \times M}$ such that:

$$s_{p,m} = \text{NA} \iff M_{p,m} = 0. \quad (4.1)$$

This missingness reflects the biological identity of the phage, so it is not missing randomly. Imputing NA values to zero would falsely treat unannotated proteins as evidence against infection which would violate the nonconformity interpretations of Section 3.3. We therefore preserve missingness by using NaN aware aggregation (Section 4.5).

Table 1: Dataset Summary and Class Distributions

<Set	Total Phages	Gram-Negative	Gram-Positive
Full Dataset	18,474	10,735	7,708
Test Split (Fold 0)	4,433	2,811	1,611
Validation Split (Fold 1)	3,173	—	—
Training Splits (Folds 2–4)	10,868	7,924	6,097

4.2 Protein to Phage Aggregation

- The raw data is at the protein level but conformal prediction operates at the phage level — aggregation is needed
- NaN-aware mean pooling: for each phage and each functional score column, compute the mean across all proteins of that phage that have a non-NaN value for that column. Proteins with NaN for a given column are excluded from the mean for that column only
- Why not replace NaN with zero before pooling: treating an unannotated protein as having zero evidence would push the phage’s aggregated score artificially toward zero, misrepresenting its biology
- After aggregation: result is a matrix of shape (number of phages) $\times$ 40, still with NaNs in cells where none of the phage’s proteins were annotated with that function

4.2.1 NaN handling after aggregation

- After mean pooling, some phages still have NaN in certain columns -> this happens when none of their proteins were annotated with a particular function
- Strategy: fill remaining NaNs using column-wise medians computed strictly on the training set (never the test set, to prevent data leakage) - The median cap: any training median above 0.6 is capped at 0.5 -> this prevents an imputed value from falsely pushing a phage toward a particular class - any remaining NaNs after median imputation are filled with 0.5 (neutral value)

4.3 Protein-to-Phage Aggregation via NaN-Aware Mean Pooling

Because predictions are produced at the individual protein level $p \in \mathcal{P}_X$, whereas host infection occurs at the intact virion level X , protein scores must be aggregated to form a

single K -dimensional score vector $\mathbf{s}(X)$ per phage accession 4.

For a phage X consisting of M_X proteins $\{p_1, \dots, p_{M_X}\}$, the aggregated score $s_k(X)$ for functional category $k \in \{1, \dots, K\}$ is computed via dimension-wise, NaN-aware mean pooling 4:

$$s_k(X) = \begin{cases} \frac{\sum_{m=1}^{M_X} P_{p_m,k} \cdot \mathbf{1}(P_{p_m,k} \neq \text{NA})}{\sum_{m=1}^{M_X} \mathbf{1}(P_{p_m,k} \neq \text{NA})} & \text{if } \sum_{m=1}^{M_X} \mathbf{1}(P_{p_m,k} \neq \text{NA}) > 0, \\ \text{NA} & \text{otherwise.} \end{cases} \quad (4.2)$$

Correspondingly, the missingness mask element $m_k(X)$ is set to 1 if at least one protein in phage X possesses a valid score for category k , and 0 otherwise 4. This pooling operation reduces the dataset to a matrix $\mathbf{S} \in ([0, 1] \cup \{\text{NA}\})^{18474 \times K}$ paired with mask matrix $\mathbf{M} \in \{0, 1\}^{18474 \times K}$, avoiding global score fabrication 4.

4.4 Protein-to-Phage Aggregation and NaN Handling

Host-range prediction is performed at the phage level, so protein-level scores must be aggregated. For phage p with protein set $\mathcal{T}_p$ and score dimension m , the NaN-aware mean is

$$\bar{s}_{p,m} = \frac{1}{|\mathcal{T}_{p,m}|} \sum_{t \in \mathcal{T}_{p,m}} s_{t,m}, \quad \mathcal{T}_{p,m} = \{t \in \mathcal{T}_p : s_{t,m} \neq \text{NA}\}. \quad (4.3)$$

If $\mathcal{T}_{p,m} = \emptyset$ then $\bar{s}_{p,m} = \text{NA}$. This occurs when none of the phage's proteins were annotated with function m . The aggregation preserves the structural missingness pattern: a phage lacking tail-fiber proteins receives NA in the tail-fiber dimension, rather than a false zero.

Post-aggregation imputation. After pooling, some phages still have NA in certain dimensions because no protein carried that annotation. These are filled by column-wise medians computed strictly on the training set. Let $\mathcal{I}_{\text{train}}$ denote the training phage indices. For dimension m ,

$$\mu_m = \text{median}(\{\bar{s}_{p,m} : p \in \mathcal{I}_{\text{train}}, \bar{s}_{p,m} \neq \text{NA}\}). \quad (4.4)$$

Any training median above 0.6 is capped at 0.5 to prevent imputed values from falsely pushing a phage toward a particular class. Remaining NA values after median imputation are filled with 0.5, the neutral midpoint of the score range. This two-stage imputation (median then neutral) is applied after aggregation and only to the training set statistics, avoiding data leakage from test to train.

Resulting phage-level matrix. After aggregation and imputation, each phage p is represented by a pair $(\bar{\mathbf{s}}_p, \mathbf{m}_p) \in [0, 1]^M \times \{0, 1\}^M$, where $\mathbf{m}_p$ records which dimensions

were originally observed ($m_{p,k} = 1$) versus imputed ($m_{p,k} = 0$). This pair is the input to the conformal pipeline of Chapter 3.

4.5 Protein-to-Phage Aggregation (NaN-Aware Mean Pooling)

Because base classifiers predict functional scores at the individual protein level, whereas host recognition is an emergent property of the entire viral particle, protein-level predictions must be aggregated into a unified phage-level representation.

Let $\mathcal{P}_i = \{p_{i,1}, p_{i,2}, \dots, p_{i,n_i}\}$ denote the set of n_i constituent proteins belonging to bacteriophage accession $i \in \{1, \dots, N_{\text{phage}}\}$, where $N_{\text{phage}} = 18474$. For each functional dimension $m \in \{1, \dots, M\}$, the aggregated phage-level raw score $s_{i,m}$ is computed using dimension-wise, NaN-aware mean pooling:

$$s_{i,m} = \begin{cases} \frac{\sum_{p \in \mathcal{P}_i} s_{p,m} \cdot \mathbf{1}(M_{p,m} = 1)}{\sum_{p \in \mathcal{P}_i} \mathbf{1}(M_{p,m} = 1)} & \text{if } \sum_{p \in \mathcal{P}_i} M_{p,m} > 0, \\ \text{NA} & \text{if } \sum_{p \in \mathcal{P}_i} M_{p,m} = 0. \end{cases} \quad (4.5)$$

Equation (4.5) excludes missing entries (NA) from both the numerator and denominator, averaging strictly over observed proteins. A phage-level score $s_{i,m}$ evaluates to NA if and only if none of the phage’s constituent proteins were assigned a prediction for functional category m .

Correspondingly, the aggregated phage-level binary mask vector $\mathbf{m}_i = (m_{i,1}, \dots, m_{i,M})^T \in \{0, 1\}^M$ is defined coordinate-wise as:

$$m_{i,m} = \mathbf{1} \left(\sum_{p \in \mathcal{P}_i} M_{p,m} > 0 \right). \quad (4.6)$$

This operation reduces the protein score tensor to a phage-level score matrix $\mathbf{S} \in ([0, 1] \cup \{\text{NA}\})^{N_{\text{phage}} \times M}$ paired with a phage-level missingness mask $\mathbf{M} \in \{0, 1\}^{N_{\text{phage}} \times M}$.

As established in Chapter 3, our conformal envelope constructions (Radial, Strip, and Collapsed 1D) natively operate on the paired observation $(\mathbf{s}_i, \mathbf{m}_i)$, requiring no prior score imputation. However, for diagnostic baseline comparison or when evaluating downstream classifiers that demand dense numeric arrays, post-aggregation missing values are imputed strictly on the training fold D_{train} (folds 2–4) to avoid data leakage:

- (i) **Training Median Computation:** For each dimension m , the median observed value $\tilde{\mu}_m = \text{Median}(\{s_{i,m} : i \in D_{\text{train}}, m_{i,m} = 1\})$ is evaluated exclusively on training phages.
- (ii) **Neutral Cap Enforcement:** To prevent imputed fallback values from falsely inflating evidence toward a particular class hypothesis, any column median exceeding

0.6 is capped at 0.5:

$$\tilde{\mu}_m^{\text{capped}} = \min(\tilde{\mu}_m, 0.5). \quad (4.7)$$

- (iii) **Imputation:** Remaining NA entries across validation and test folds are replaced using $\tilde{\mu}_m^{\text{capped}}$ (or 0.5 as a neutral default if no training observations exist for feature m).

All main conformal classification results reported in Chapter 5 rely entirely on the native missingness pair $(\mathbf{s}_i, \mathbf{m}_i)$ without applying post-aggregation imputation.

4.6 Train-Test Split

- Standard split: split 0 is the held-out test set, splits 1–4 form the training set
- The contamination problem: explain the protein cluster (pc) column -> a protein's cluster may include proteins from phages in different folds. The models that produced the scores were trained on protein clusters, so a protein from a training-fold phage may share a cluster with a test-fold protein, introducing subtle leakage
- The masking fix: the file `all_random_pc_splits.pkl` encodes for each protein and each (host, function) model which fold that protein was assigned to. Setting scores to NaN where the protein was not in the test set of the relevant model, before aggregation, removes the contaminated predictions. The existing NaN-aware mean pooling then handles these naturally -> this is the "missing value" framing

4.7 Data Partitioning and the Contamination Fix

To guarantee finite-sample conformal validity (Theorem 2.3), calibration samples S_2 and test samples S_3 must be exchangeable 4. In biological sequence classification, naive random splitting introduces *data contamination* due to homologous protein clusters shared across distinct phage accessions 4, 5.

4.7.1 Protein Cluster Contamination Problem

Base scoring models were trained at the protein cluster level (pc)[cite: 5]. If proteins from the same cluster appear in both the training set of a base scoring model and a test phage, the predicted scores $s_k(X_{\text{test}})$ become optimistically biased, violating exchangeability 4. For example, in the host genus *Ralstonia*, 4 out of 46 test protein clusters (8.7%) overlap with training clusters, introducing score leakage 5.

4.7.2 Protein-Level Masking Strategy

To eliminate optimistic bias, we enforce a strict 5-fold cross-validation partition based on genome clusters 4, 5:

- **Split 0 (Test Fold):** Consists of 4,433 phages reserved strictly for final holdout evaluation 4, 5.
- **Split 1 (Validation Fold):** Consists of 2,593 phages used exclusively to tune envelope hyperparameters (e.g., bin count NB , directional resolution M , and kernel smoothing κ) 5.
- **Splits 2–4 (Training/Calibration Folds):** Consists of 9,101 phages used to discover envelope boundaries (S_1) and calibrate scaling factors (S_2) 5.

Before phage-level mean pooling via Equation (4.2), protein-level prediction matrices are masked: any protein score that was not in the designated evaluation fold of the base model is set to NA 4. The NaN-aware mean pooling operator inherently filters these masked entries out, ensuring that all scores entering the conformal calibration pipeline are unbiased by construction 4.

5 Experiments and Results

5.1 Synthetic Data Validation

5.1.1 Purpose and setup

- Why synthetic validation before real data: allows controlled verification that the coverage guarantee holds exactly, with known ground truth distributions
- General setup: simulate score vectors from known distributions, apply all three envelope methods, verify empirical coverage matches theoretical target $1 - \alpha$

5.1.2 2D Experiments

- Three scenarios tested in 2D ($K=2$):
 - Uncorrelated scores: two independent uniform or Gaussian dimensions
 - Correlated scores: scores with positive correlation -> tests whether the envelope adapts to the correlation structure
 - Asymmetric predictors: one informative dimension and one uninformative dimension -> tests whether the envelope correctly ignores the noisy dimension
- For each scenario: show the envelope shape visually, report empirical coverage across multiple runs, confirm it matches $1 - \alpha$
- Key observation from 2D: the radial and strip envelopes are visibly non-convex and wrap tightly around the data, while collapsed 1D produces a simple interval

5.1.3 Extension to K Dimensions

- Repeat the experiments for $K \in 2, 5, 10, 20$ with a single host
- Show that all three methods maintain coverage $\geq 1 - \alpha$ across all dimensionalities
- Discuss the effect of increasing K on envelope quality: radial method becomes sparser (fewer S1 points per sector), strip method has more dimension pairs to condition on, collapsed 1D is unaffected
- Show that the analytical τ computation gives identical results to what iterative search would give, confirming correctness

5.1.4 Multiple Hosts

- Extend to a simulated multi-class setting with multiple hosts ($|\mathcal{Y}| = 6, K = 5$)
- Each host's true score distribution is a distinct cluster in $[0, 1]^K$
- Verify that the single unified envelope correctly assigns the true host to the prediction set with probability $\geq 1 - \alpha$
- Show 3D visualisation of the envelope for $K=3$

5.2 Gram Type Classification

5.2.1 Experimental Setup

- Data: split-0 phages only (80% calibration, 20% test), unknown host types excluded
- Parameters: $\alpha = 0.1$, $M = 200$ directions (radial), $M=20$ bins (strip), 10 independent runs to estimate variance
- The NC transformation used: keep scores as-is for one class, flip with $1 - s$ for the other

5.2.2 Results

- table: overall coverage, per-class coverage, average set size, empty set rate, size-2 rate, singleton accuracy std across 10 runs for all three methods
- Bar charts: set size distribution, overall and per-class coverage

5.3 Host Classification

6 Conclusion and Future Work

6.1 Summary.....

6.2 Limitations

- gram-pos coverage -

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